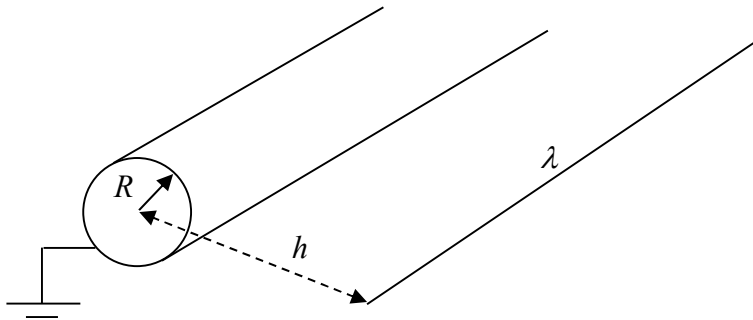


PHL1010: Practice Problems
(10/10/2025)

- Problems from Griffiths (3rd edition)
2.16, 2.17, 2.18, 2.27, 2.38, 2.47, 3.14, 3.34, 3.49, 4.5, 4.6, 4.10, 4.15, 4.16, 4.28, 5.10, 5.13, 5.16
- Consider a line charge λ at a distance h from the center of a conducting cylinder of radius R . Cylinder is grounded (see the figure). Find the potential at a point $r > R$ using method of images.



- (a) Using the law of cosines, show that the Eq. 3.17 (Griffiths, 3rd edition, pp. 125) can be written as following:

$$V(r, \theta) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{r^2 + a^2 - 2ra \cos \theta}} - \frac{q}{\sqrt{R^2 + (ra/R)^2 - 2ra \cos \theta}} \right]$$

where, r and θ are the usual polar coordinates with the z -axis along the line through q .

Note: this form makes it obvious that $V = 0$ on the sphere.

- Find the induced surface charge on the sphere. Using this expression, plot the induced charge density assuming the point charge is kept (i) a distance twice that of the radius of the conducting sphere, (ii) a distance four times that of the radius of the conducting sphere by calculating V at $\theta = 0, \pi/2$ and at π . What important information you can draw from this plot?
 - Give a rough sketch for the line of forces in the region surrounding this charge system.
 - Consider a charge system comprising an insulating conducting sphere (radius a) having a charge Q on it, and a point charge q kept at a distance d ($> a$) from the center of this sphere. Using the Method of Images, find potential at a point $\vec{r}$ (where $|\vec{r}| > a$).
- Two infinitely extended conducting cones ($\theta = \pi/3$ and $\pi/6$) are separated by an insulating infinitesimal gap at $r = 0$ (see the figure). If $V(\theta = \pi/3) = 0$, and $V(\theta = \pi/6) = 100V$, find V and $|\vec{E}|$ between the cones.

